

# EECE 430 — Software Engineering

## Final Examination — Practice Exam 2

---

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Student ID: \_\_\_\_\_

Section: \_\_\_\_\_

### Instructions:

- This exam contains **33 multiple-choice questions**.
  - Time allowed: **1 hour**.
  - Circle the letter of the **best** answer for each question.
  - Each question is worth equal marks.
  - No electronic devices, notes, or textbooks are permitted.
- 

**1. The Agile Manifesto states that while there is value in processes and tools, which of the following is valued MORE?**

- A) Comprehensive documentation
- B) Contract negotiation
- C) Individuals and interactions
- D) Following a plan

**2. A developer practicing TDD writes a test, runs it, and it passes immediately without writing any new code. What does this most likely indicate?**

- A) The TDD process is working perfectly
- B) The functionality already exists or the test is flawed and does not properly validate the new requirement
- C) The developer should move on to refactoring
- D) The test framework has a bug

**3. Which of the following is TRUE about agile method applicability?**

- A) Agile is best suited for large systems with strict regulatory requirements
- B) Agile works well for small/medium-sized products and custom systems where the customer is committed to involvement
- C) Agile requires that all requirements be fully specified before development begins
- D) Agile eliminates the need for any software architecture planning

**4. What is the primary purpose of Lo-Fi UI sketches and storyboards in agile development?**

- A) To replace the need for any user interface coding
- B) To create final production-ready designs
- C) To get customer participation in UI design and show interactivity without building a prototype, since they are easier to change than code
- D) To document the system architecture

**5. In agile estimation using the Fibonacci scale, a user story estimated at 25 or more points should typically be:**

- A) Assigned to the most experienced developer
- B) Divided into simpler, smaller user stories
- C) Moved to the bottom of the product backlog
- D) Estimated again using a linear scale instead

**6. Which of the following BEST describes the role of the Product Owner in Scrum?**

- A) Writing code and conducting code reviews
- B) Identifying product features, prioritizing them, and continuously reviewing the product backlog
- C) Facilitating daily stand-up meetings and removing impediments

D) Deciding the technical architecture of the system

**7. A Scrum team discovers during sprint execution that a critical PBI cannot be completed. According to Scrum principles, what should happen?**

- A) The sprint should be extended by one week
- B) The ScrumMaster should complete the PBI personally
- C) The unfinished PBI is returned to the product backlog and queued for a future sprint
- D) The entire sprint is cancelled and restarted

**8. In XP, what does 'sustainable pace' mean?**

- A) Developers should work at maximum speed at all times
- B) Large amounts of overtime are not acceptable as the net effect reduces code quality and medium-term productivity
- C) Each sprint must be completed faster than the previous one
- D) Only senior developers should set the pace of development

**9. Which of the following correctly lists the four product backlog activities?**

- A) Planning, coding, testing, deploying
- B) Refinement, estimation, prioritization, creation
- C) Sprint planning, sprint execution, sprint review, sprint retrospective
- D) Requirements gathering, design, implementation, maintenance

**10. Why does Scrum recommend an ideal team size of 5-8 people?**

- A) Larger teams require less management overhead
- B) It is large enough to be diverse in skills yet small enough to communicate informally and effectively
- C) Smaller teams cannot handle any meaningful workload
- D) It matches the number of Scrum ceremonies

**11. Which architectural design issue arises when a system needs to support a rapidly changing number of users?**

- A) Software reuse
- B) Software compatibility
- C) The architecture must be designed so the system can be scaled up and down
- D) Product lifetime

**12. A system requires both high availability and redundancy. Which architectural approach addresses this, and what is its trade-off?**

- A) Layered architecture — trade-off is reduced security
- B) Including redundant components and fault tolerance mechanisms — trade-off is increased development cost, complexity, and potential for bugs
- C) Pipe and filter — trade-off is slower data processing
- D) MVC pattern — trade-off is more complex user interfaces

**13. In the MVC pattern, what happens when the data in the Model changes?**

- A) The Controller must be rewritten to handle the new data format
- B) The change can be reflected in all Views, supporting multiple presentations of the same data
- C) The View must be completely redesigned
- D) The Model is discarded and a new one is created

**14. A transaction processing system handles user requests to query and update a database. According to the course, which component acts as an intermediary between user input and the database?**

- A) The pipe and filter component
- B) The transaction manager, which sits between the application processing logic and the database
- C) The API gateway
- D) The service registry

**15. Which statement about cross-cutting concerns in a layered architecture is CORRECT?**

- A) Cross-cutting concerns only affect the top layer of the architecture
- B) Security is NOT a cross-cutting concern because it only applies to the database layer
- C) Cross-cutting concerns affect all layers, creating interactions between them, which is why adding security post-design is difficult
- D) Cross-cutting concerns can be eliminated by using the MVC pattern

**16. What is the key difference between IaaS and PaaS?**

- A) IaaS provides complete software applications; PaaS provides only hardware
- B) IaaS provides infrastructure services like compute, network, and storage for virtual servers; PaaS provides libraries, frameworks, and database access on top of that
- C) PaaS requires you to manage the operating system; IaaS does not
- D) IaaS and PaaS are identical but offered by different cloud vendors

**17. In a multi-tenant SaaS system, a software bug causes data belonging to Company A to become visible to Company B. This vulnerability is a consequence of:**

- A) Using a multi-instance database architecture
- B) Logical isolation via tenant identifiers rather than physical isolation, meaning a single bug can expose all tenants' data
- C) Encrypting data at rest
- D) Using container-based deployment

**18. Which of the following is a benefit of SaaS for the SOFTWARE PROVIDER (not the customer)?**

- A) No upfront costs for software or servers
- B) Mobile, laptop, and desktop access
- C) Update management — the provider controls updates and all customers receive them simultaneously
- D) Reduced software management costs for the customer

**19. A company wants to avoid vendor lock-in and ensure their cloud application can survive the failure of a single cloud provider. Which cloud deployment model should they choose?**

- A) Private Cloud
- B) Public Cloud
- C) Hybrid Cloud
- D) Multi-Cloud

**20. What is the PRIMARY advantage of using containers over VMs for implementing cloud-based multi-instance systems?**

- A) Containers provide stronger hardware-level isolation
- B) Containers run different operating systems more effectively
- C) Containers are much smaller (megabytes vs gigabytes), start in seconds, and use fewer resources since they share the host OS kernel
- D) Containers are more secure because each has its own hypervisor

**21. Which of the following is NOT one of the three core DevOps principles?**

- A) Everyone is responsible for everything
- B) Everything that can be automated should be automated
- C) Separate teams should handle development, testing, and operations independently
- D) Measure first, change later

**22. In Git, a developer creates a branch, makes changes, and wants to merge back to the master. Both the developer and another team member have modified the same lines in the same file. What happens?**

- A) Git automatically keeps the most recent change
- B) Git signals a merge conflict and the developers must resolve it together
- C) Git rejects the merge permanently and the branch must be deleted
- D) Git creates a third version that combines both changes automatically

**23. What problem does continuous delivery solve that continuous integration alone does not?**

- A) Continuous delivery runs unit tests, which CI does not
- B) Continuous delivery tests the system in a production-replica environment, catching bugs that only appear in the real operational environment
- C) Continuous delivery replaces the need for version control
- D) Continuous delivery eliminates the need for automated testing

**24. Configuration management tools like Puppet and Chef are used in Infrastructure as Code to:**

- A) Write application source code automatically
- B) Automatically install software and services on servers according to the infrastructure definition, and push updates consistently
- C) Replace the need for a code management system like Git
- D) Monitor application performance in production

**25. A/B testing, as described in the context of continuous deployment, involves:**

- A) Testing two completely different applications against each other
- B) Deploying a new version on some servers while keeping the old version on others, then comparing how customers use each
- C) Running automated tests in two separate environments simultaneously
- D) Having two developers independently test the same feature

**26. Why are ACID transactions considered impractical in a microservices architecture?**

- A) Microservices do not use databases
- B) ACID transactions require bundling updates into a single unit, which is impractical unless all data is confined to a single microservice
- C) ACID transactions are too fast for microservices to handle
- D) Microservices only support NoSQL databases which do not support any transactions

**27. A photo printing service uses an API gateway between the mobile app and its microservices. Which of the following is NOT a benefit of using an API gateway?**

- A) The app does not need to know what communication protocol the microservices use
- B) It is possible to split or combine services without affecting the client app
- C) It eliminates the need for authentication in individual microservices
- D) It acts as a single point of contact that translates app requests into microservice calls

**28. Which decomposition guideline states that elements likely to be changed at the same time should be located within the same microservice?**

- A) Single responsibility principle
- B) Common closure principle
- C) Business capability association
- D) Data isolation principle

**29. A microservices system experiences an external service failure where a critical service becomes unresponsive. The simplest way to report this type of failure to calling services is by using:**

- A) Custom binary error protocols
- B) Email notifications to the development team
- C) HTTP status codes indicating whether the request succeeded or failed
- D) Writing error messages to a shared log file that all services monitor

**30. A user clicks a link in a phishing email that leads to a fake website designed to collect their login credentials. This type of attack exploits which weakness of password-based authentication?**

- A) Insecure passwords
- B) Password reuse
- C) Phishing attacks — users are tricked into entering credentials on fraudulent sites
- D) Forgotten password recovery mechanisms

**31. A Key Management System (KMS) must maintain multiple timestamped versions of encryption keys. What is the primary reason for this requirement?**

- A) To allow different users to use different keys simultaneously
- B) Because encryption keys should be changed regularly, and archived data encrypted with older keys must still be decryptable
- C) To speed up the encryption process by rotating keys
- D) To comply with open-source licensing requirements

**32. Which combination of defensive measures BEST protects against session hijacking?**

- A) Long session timeouts and unencrypted HTTP connections
- B) Traffic encryption (HTTPS), short session timeouts, and multi-factor authentication
- C) Disabling cookies entirely and using only GET requests
- D) Allowing unlimited login attempts and storing passwords in plain text

**33. According to the GDPR requirements discussed in the course, software companies must:**

- A) Hide their privacy policy in a long terms-and-conditions document
- B) Include a summary of their privacy policy written in plain language on their website
- C) Only collect data from users who are software engineers
- D) Store all user data indefinitely regardless of account status

---

— END OF QUESTIONS —

## EECE 430 — Practice Exam 2 — Answer Key

---

### Quick Reference:

| Q   | Ans | Q   | Ans | Q   | Ans |
|-----|-----|-----|-----|-----|-----|
| 1.  | C   | 2.  | B   | 3.  | B   |
| 4.  | C   | 5.  | B   | 6.  | B   |
| 7.  | C   | 8.  | B   | 9.  | B   |
| 10. | B   | 11. | C   | 12. | B   |
| 13. | B   | 14. | B   | 15. | C   |
| 16. | B   | 17. | B   | 18. | C   |
| 19. | D   | 20. | C   | 21. | C   |
| 22. | B   | 23. | B   | 24. | B   |
| 25. | B   | 26. | B   | 27. | C   |
| 28. | B   | 29. | C   | 30. | C   |
| 31. | B   | 32. | B   | 33. | B   |

### Detailed Explanations:

- 1. C** — The Agile Manifesto explicitly values individuals and interactions over processes and tools.
- 2. B** — In TDD, a new test should fail initially. If it passes without new code, either the functionality already exists or the test is not correctly checking the new requirement.
- 3. B** — Agile is suited for small/medium products for sale and custom systems with customer commitment and few external regulations.
- 4. C** — Lo-Fi UI sketches help customers participate in design. They are less intimidating, easier to change, and focus on interaction rather than visual polish.
- 5. B** — Stories estimated at 25+ are too complex. The team should divide them into simpler stories that can be estimated and implemented more reliably.
- 6. B** — The Product Owner identifies features, prioritizes requirements, and continuously reviews the product backlog.
- 7. C** — Sprints are timeboxed and never extended. Unfinished items go back to the product backlog for future sprints.
- 8. B** — Sustainable pace means no excessive overtime, as it reduces code quality and productivity in the medium term.
- 9. B** — The four backlog activities are: refinement (analyze/refine PBIs), estimation (assess effort), prioritization (reorder), and creation (add new items).
- 10. B** — 5-8 is large enough for skill diversity but small enough for effective informal communication. Teams are self-organizing and take collective responsibility.
- 11. C** — When user numbers change quickly, the architecture must support scaling. This is the 'number of users' design issue.
- 12. B** — Availability requires redundant components and fault tolerance. The trade-off is added complexity, cost, and increased chance of bugs/vulnerabilities.
- 13. B** — MVC allows data to change independently of representation. Changes in one representation are shown in all Views.

- 14. B** — Transaction processing systems use: I/O processing -> application logic -> transaction manager -> database.
- 15. C** — Cross-cutting concerns like security affect ALL layers. Distributing them across layers creates inter-layer dependencies, making post-design changes difficult.
- 16. B** — IaaS provides infrastructure (compute, network, storage). PaaS adds libraries, frameworks, and services like databases on top of infrastructure.
- 17. B** — Multi-tenant systems use logical isolation (tenant IDs). A single bug or breach can expose all tenants' data — this is a known disadvantage.
- 18. C** — Update management is a provider benefit: they control updates, all customers get them at the same time, reducing maintenance of multiple versions.
- 19. D** — Multi-cloud uses multiple cloud providers, avoiding data lock-in and single points of failure. But you must handle API differences and cross-cloud migration.
- 20. C** — Containers are smaller, start faster, and share the host OS kernel, making them more resource-efficient than VMs.
- 21. C** — DevOps eliminates separate teams. The three principles are: everyone responsible for everything, automate everything, measure first change later.
- 22. B** — When the same lines are changed in different branches, Git signals a conflict. Developers must collaborate to resolve it manually.
- 23. B** — CI builds and tests on the dev machine. CD additionally tests in a production-replica (staged) environment, catching environment-specific bugs.
- 24. B** — CM tools like Puppet/Chef read infrastructure models and automatically provision servers, install software, and push changes consistently.
- 25. B** — A/B testing deploys new version on some servers, old on others, and diverts customers to compare feature usage and effectiveness.
- 26. B** — ACID bundles updates into one atomic unit. This is impractical across microservices since each has its own database — you'd need distributed transactions.
- 27. C** — The API gateway does NOT eliminate the need for authentication in individual services. It insulates the client, handles routing, and enables service decomposition changes.
- 28. B** — The common closure principle: elements that change together should be in the same service, so most new requirements affect only one service.
- 29. C** — The simplest way to report microservice failures is using HTTP status codes, which indicate success or failure of requests.
- 30. C** — Phishing attacks trick users into clicking fake links and entering credentials on fraudulent sites that mimic legitimate ones.
- 31. B** — Keys are changed regularly for security. Archived data may use older keys, so KMS must keep timestamped versions to decrypt backups/archives when needed.
- 32. B** — HTTPS prevents traffic monitoring for cookies, short timeouts limit the window of attack, and MFA confirms identity for sensitive actions.
- 33. B** — GDPR requires a plain-language summary of the privacy policy on the website, not buried in long legal documents that nobody reads.

---

— END OF ANSWER KEY —